Constrained Vortex-Fluid Action: A Technical Repair of VAM-5

SST Lagrangian, Hamiltonian and finite-core audit

Replacement for the mathematical claims in Sections VI–IX and related benchmarks

Mathematical reconstruction of the historical manuscript by O. Iskandarani

Original VAM-5: doi:10.5281/zenodo.15772858

20 October 2026 | Version 0.1.0 | **DEVELOPMENT / NOT PHYSICALLY CERTIFIED**

Abstract

The 2025 VAM-5 manuscript proposes a vorticity/helicity Lagrangian intended to reconstruct gravity, electromagnetism, quantum phases and particle masses. Its Euler–Lagrange and Hamiltonian calculations do not follow from the stated field variables; some energy-density terms are dimensionally inconsistent; and a globally potential velocity field cannot represent a smooth, nonzero-vorticity core. This technical repair installs a mathematically valid constrained variational model of incompressible Euler flow, its canonical Hamiltonian in material coordinates, a separately labeled *optional* Euler–Poincaré H^1-regularized model, and a boundary between hydrodynamic results and still-unverified SST interaction hypotheses. All examples are algebraic or finite-grid regression tests; no PKLSA knot has been dynamically simulated or certified here.

Contents

1 Scope, sources and scientific status **2**

2 Canonical notation and dimension audit **2**

3 Exact constrained incompressible Euler action **3**

4 Correct canonical Hamiltonian and its interpretation **4**

5 Optional H^1 finite-core regularization: separate model **4**

6 Helicity, field mapping and topology: explicit non-results **5**

7 Phase observables, vortex clocks and non-circular tests **5**

8 Numerical scale check with canonical SST constants **6**

9 Verification contract and decision gates **6**

10 Minimal patch instructions for a future VAM-5 revision **7**

1 Scope, sources and scientific status

This is a *new replacement document*, not an in-place edit of the author's original 59-page PDF. The old VAM-5 text remains the historical statement. The present manuscript changes the mathematical assertions summarized below, while preserving the research motivation of a flat three-dimensional incompressible, inviscid medium and vortex-knot excitations.

Old location	Original mathematical issue	Corrective disposition
pp. 10–13; (23)–(31)	Euler equations presented as variations of a velocity-only functional; dimensional mismatch	Replace with constrained material-map action (Section 3); label EM and gravity as unresolved.
pp. 14–16; (33)–(38)	$\partial\mathcal{L}/\partial\dot{\mathbf{v}}$ given nonzero although $\dot{\mathbf{v}}$ is absent	Use canonical momentum conjugate to the material position $\mathbf{X}$, not to an arbitrary velocity field.
pp. 17–19; (39)–(49)	Globally potential velocity with nonzero curl; phase treated as both dimensionless clock and action	Separate physical vortex phase observables from Hamilton–Jacobi action and from circulation quantization.
pp. 3–9; (1)–(22)	Unclosed clock, helicity-charge and gravitational identifications	Preserve as distinct falsifiable hypotheses without presenting them as field equations.
pp. 38–55; benchmarks	Energy presented as mass and fitted factors treated as predictions	Require energy integrals, E/c^2 and independent parameter closure.

Source: Iskandarani (2025), original VAM-5 PDF, pp. 1–59 [1].

2 Canonical notation and dimension audit

We use absolute Euclidean time t, spatial position $\mathbf{x}$, fluid velocity $\mathbf{v}(\mathbf{x},t)$, vorticity $\boldsymbol{\omega} = \nabla \times \mathbf{v}$ and incompressible background mass density ρ_f. The canonical SST figures are

$$|\mathbf{v}_{\circlearrowleft}| = 1.09384563 \times 10^{6}\ \mathrm{m\,s^{-1}}, \qquad r_c = 1.40897017 \times 10^{-15}\ \mathrm{m}, \tag{1}$$

$$\rho_f = 7.0 \times 10^{-7}\ \mathrm{kg\,m^{-3}}, \qquad \rho_{\mathrm{core}} = 3.8934358266918687 \times 10^{18}\ \mathrm{kg\,m^{-3}}. \tag{2}$$

Here ρ_E is an *energy density* ($\mathrm{J\,m^{-3}}$) and $\rho_m = \rho_E/c^2$ is a *mass-equivalent density* ($\mathrm{kg\,m^{-3}}$). The background density and an assumed core density must not be substituted for one another without a constitutive model.

Dimensional requirements: $[\mathbf{v}] = \mathrm{m\,s^{-1}}$, $[\boldsymbol{\omega}] = \mathrm{s^{-1}}$, $[p] = [\mathcal{L}] = \mathrm{J\,m^{-3}}$. Hence

$$\left[\rho_f|\mathbf{v}|^2\right] = \mathrm{J\,m^{-3}}, \tag{3}$$

$$\left[\rho_E|\boldsymbol{\omega}|^2\right] = \mathrm{J\,m^{-3}\,s^{-2}}, \tag{4}$$

$$\left[(\mathbf{v}\cdot\boldsymbol{\omega})^2\right] = \mathrm{m^2\,s^{-4}}. \tag{5}$$

Consequently the original coefficient of $\rho_E|\boldsymbol{\omega}|^2$ needs $[\lambda_g] = \mathrm{s^2}$ and the coefficient of local $(\mathbf{v}\cdot\boldsymbol{\omega})^2$ needs $[\alpha_e] = \mathrm{kg\,s^2\,m^{-3}}$ if the respective terms are energy densities. The ordinary dimensionless electromagnetic fine-structure constant α cannot itself fulfill the second role. The repaired fluid theory uses no such gravitational or electromagnetic coefficients.

3 Exact constrained incompressible Euler action

3.1 Material coordinates and incompressibility

Choose fixed reference labels $\mathbf{a} \in \Omega_0$ and a time-dependent fluid-particle diffeomorphism $\mathbf{X}(\mathbf{a}, t)$. Define its deformation gradient and Jacobian by

$$F_{iA} = \frac{\partial X_i}{\partial a_A}, \qquad J = \det F, \qquad \mathbf{v}(\mathbf{X}(\mathbf{a}, t), t) = \partial_t \mathbf{X}(\mathbf{a}, t). \tag{6}$$

Constant reference density ρ_f and $J = 1$ imply spatial incompressibility. The correct material-map action (no external forcing) is

$$\boxed{\mathcal{A}_{\text{Euler}}[\mathbf{X}, p] = \int_{t_0}^{t_1} \mathrm{d}t \int_{\Omega_0} \mathrm{d}^3 a \left\{ \frac{\rho_f}{2} |\partial_t \mathbf{X}|^2 + p(\mathbf{a}, t)(J - 1) \right\}.} \tag{7}$$

Here $p(\mathbf{a}, t)$ is the pullback of a spatial pressure field, serving as a Lagrange multiplier. Variations vanish at the time endpoints and obey periodic or appropriate impermeable-boundary conditions.

3.2 Explicit variation

At fixed labels, the determinant variation and Piola identity read

$$\delta J = J(F^{-T})_{Ai} \frac{\partial \delta X_i}{\partial a_A}, \qquad \partial_{a_A} \left[J(F^{-T})_{Ai} \right] = 0. \tag{8}$$

Integrating by parts in time and labels gives

$$\delta_{\mathbf{X}} \mathcal{A}_{\text{Euler}} = \int \mathrm{d}t \mathrm{d}^3 a \left[-\rho_f \partial_t^2 X_i - \partial_{a_A} (p J (F^{-T})_{Ai}) \right] \delta X_i, \tag{9}$$

$$\delta_p \mathcal{A}_{\text{Euler}} = \int \mathrm{d}t \mathrm{d}^3 a \, (J - 1) \, \delta p. \tag{10}$$

At $J = 1$, the spatial form of these Euler–Lagrange equations is

$$\boxed{\rho_f \left(\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right) = -\nabla p, \qquad \nabla \cdot \mathbf{v} = 0.} \tag{11}$$

There is no arbitrary "swirl gravity" source in this exact result. Pressure enforces incompressibility and its gradients accelerate *fluid parcels*; an external test object's acceleration needs a separately specified interaction law. The standard variational structure and Hamiltonian framework are discussed in [2, 3].

3.3 Vorticity, Kelvin circulation, pressure and energy

For smooth Euler solutions under suitable boundaries, the pressure satisfies

$$\boxed{\nabla^2 p = -\rho_f \, \partial_i v_j \, \partial_j v_i.} \tag{12}$$

Taking the curl of Eq. (11) gives

$$\partial_t \boldsymbol{\omega} = \nabla \times (\mathbf{v} \times \boldsymbol{\omega}), \qquad \nabla \cdot \boldsymbol{\omega} = 0. \tag{13}$$

The Euler kinetic Hamiltonian and inviscid circulation theorem are

$$E_{\text{E}} = \frac{\rho_f}{2} \int_{\Omega} |\mathbf{v}|^2 \mathrm{d}^3 x, \tag{14}$$

$$\Gamma_{C(t)} = \oint_{C(t)} \mathbf{v} \cdot \mathrm{d}\boldsymbol{\ell} = \text{constant} \quad \text{for a material loop } C(t). \tag{15}$$

For a moving uniform background in an unbounded domain, a finite *relative* energy and its boundary/impulse terms must be specified rather than integrating a spatially uniform velocity over infinite volume.

4 Correct canonical Hamiltonian and its interpretation

The canonical configuration variable is $\mathbf{X}$, not $\mathbf{v}$. In label coordinates its conjugate momentum is

$$\boxed{\mathbf{P}(\mathbf{a},t) = \frac{\delta L}{\delta(\partial_t \mathbf{X})} = \rho_f \partial_t \mathbf{X}.} \tag{16}$$

The corresponding *constrained* Hamiltonian is

$$\mathscr{H}[\mathbf{X},\mathbf{P},p] = \int_{\Omega_0} \mathrm{d}^3 a \left[\frac{|\mathbf{P}|^2}{2\rho_f} - p(J-1)\right], \quad \Rightarrow \quad \boxed{\mathscr{H}\big|_{J=1} = \frac{\rho_f}{2}\int_\Omega |\mathbf{v}|^2 \mathrm{d}^3 x.} \tag{17}$$

The pressure constraint and volume-preserving initial data must be carried through Hamiltonian evolution; writing $\mathscr{H}$ alone does not eliminate these constraints. This is the repaired counterpart of historical VAM-5 eqs. (33)–(38). A velocity-only expression with no $\dot{\mathbf{v}}$ has $\partial\mathcal{L}/\partial\dot{\mathbf{v}} = 0$ and cannot yield the claimed $\rho_f \mathbf{v}$ as its canonical time momentum.

5 Optional $\mathbf{H}^1$ finite-core regularization: separate model

A finite vortex-core radius *does not require* altering the Euler equations: a smooth finite-width vorticity distribution is already permitted by Eq. (11). The following is an independent testable regularization, *not a derivation of SST's fundamental vortex core.*

For constant length $\ell \geq 0$, define an Eulerian reduced Lagrangian with *constrained Euler–Poincaré variations*:

$$\boxed{\ell_{H^1}[\mathbf{v}] = \frac{\rho_f}{2}\int_\Omega (|\mathbf{v}|^2 + \ell^2 |\nabla \times \mathbf{v}|^2)\mathrm{d}^3 x, \quad \nabla\cdot\mathbf{v} = 0.} \tag{18}$$

The admissible velocity variations are not independent: for divergence-free displacement $\boldsymbol{\xi}$,

$$\delta\mathbf{v} = \partial_t \boldsymbol{\xi} + (\mathbf{v}\cdot\nabla)\boldsymbol{\xi} - (\boldsymbol{\xi}\cdot\nabla)\mathbf{v}. \tag{19}$$

Then, up to boundary terms,

$$\boxed{\mathbf{m} = \frac{1}{\rho_f}\frac{\delta \ell_{H^1}}{\delta \mathbf{v}} = \mathbf{v} + \ell^2 \nabla\times\boldsymbol{\omega} = \mathbf{v} - \ell^2\nabla^2\mathbf{v} \quad (\nabla\cdot\mathbf{v} = 0).} \tag{20}$$

Stationarity with these constrained variations gives

$$\boxed{\partial_t \mathbf{m} + (\mathbf{v}\cdot\nabla)\mathbf{m} + (\nabla\mathbf{v})^T\mathbf{m} = -\nabla\Pi, \qquad \nabla\cdot\mathbf{v} = 0.} \tag{21}$$

The scalar Π is a specific pressure-like multiplier with units $\mathrm{m}^2\,\mathrm{s}^{-2}$. For $\ell \to 0$, $\mathbf{m} \to \mathbf{v}$ and $(\nabla\mathbf{v})^T\mathbf{v} = \nabla(|\mathbf{v}|^2/2)$, so Eq. (21) reduces exactly to Eq. (11) after setting $p = \rho_f(\Pi + |\mathbf{v}|^2/2)$. This is the Euler–alpha family of H^1 fluid models described by Holm, Marsden and Ratiu [4, 3].

For periodic or boundary conditions permitting integration by parts, the corresponding positive energy is

$$\boxed{E_{H^1} = \frac{\rho_f}{2}\int_\Omega \mathbf{v}\cdot\mathbf{m}\,\mathrm{d}^3 x = \frac{\rho_f}{2}\int_\Omega \left(|\mathbf{v}|^2 + \ell^2|\boldsymbol{\omega}|^2\right)\mathrm{d}^3 x.} \tag{22}$$

The $\ell^2|\boldsymbol{\omega}|^2$ term is a *regularization energy*, not demonstrably an electromagnetic or gravitational field energy. A test case $\ell = r_c$ is permitted, but ℓ remains an independent free length until dynamically qualified. Standard Euler helicity conservation should not be automatically transferred to this modified model without deriving its own invariants and boundary terms.

6 Helicity, field mapping and topology: explicit non-results

Under smooth inviscid Euler flow and suitable boundary conditions, helicity

$$H = \int_\Omega \mathbf{v} \cdot \boldsymbol{\omega} \, \mathrm{d}^3x, \qquad [H] = \mathrm{m}^4\,\mathrm{s}^{-2}, \tag{23}$$

is a meaningful global invariant related to linked vortex tubes [5]. It is not electric charge: $[q] = \mathrm{C}$ and no universal conversion law is derived. In particular, an independent scaling $\mathbf{v} \mapsto a\mathbf{v}$ scales $H \mapsto a^2 H$ without necessarily changing knot type.

One may define a local scalar $h = \mathbf{v} \cdot \boldsymbol{\omega}$ with $[h] = \mathrm{m\,s}^{-2}$. Adding h^2 to a local action requires a coefficient with units $\mathrm{kg\,s^2\,m^{-3}}$ and is not generically Galilei invariant: under a uniform boost $\mathbf{v}' = \mathbf{v} + \mathbf{U}$ and $\boldsymbol{\omega}' = \boldsymbol{\omega}$ at corresponding translated coordinates,

$$h' = h + \mathbf{U} \cdot \boldsymbol{\omega}. \tag{24}$$

On a periodic domain with $\int \boldsymbol{\omega} \mathrm{d}^3x = 0$, the *global* helicity can remain invariant even though its local density changes. Thus original VAM-5 local-helicity-squared energy is not included in the repaired baseline.

Likewise, the identification $\mathbf{E} \sim \nabla \cdot \boldsymbol{\omega}$ cannot work as a vector electromagnetic field: $\nabla \cdot \boldsymbol{\omega} = 0$ identically for smooth $\boldsymbol{\omega} = \nabla \times \mathbf{v}$. Biot–Savart and the r^{-2} kernel are valuable diagnostics, but postulating an electric field as a Coulomb integral of h does not derive Maxwell's equations from the fluid dynamics. The pressure and vorticity equations alone supply neither a physical electric charge nor universal Newtonian attraction.

7 Phase observables, vortex clocks and non-circular tests

The old VAM-5 phase section introduces both a clock phase and the Hamilton–Jacobi action under the letter S. A dimensionless clock phase θ (radians) is *not* an action (joule-seconds). If $\mathbf{v} \propto \nabla S$ globally with a smooth single-valued S, then $\nabla \times \mathbf{v} = 0$ and no finite-core vorticity can be represented. A Helmholtz decomposition may contain a potential component *and* a rotational component, but this does not establish quantum mechanics or circulation quantization.

For a computed time-dependent finite-core vortex, choose a preregistered, independent observable such as a Kelvin-mode projection

$$a_k(t) = \int_\Omega \boldsymbol{\omega}(\mathbf{x}, t) \cdot \boldsymbol{\psi}_k^*(\mathbf{x}) \, \mathrm{d}^3x, \quad \theta_k(t) = \arg a_k(t), \quad \Omega_k(t) = \frac{\mathrm{d}}{\mathrm{d}t} \operatorname{unwrap} \theta_k(t), \tag{25}$$

with reference mode $\boldsymbol{\psi}_k$ independent of target relativistic formulas and $|a_k(t)|$ bounded away from zero in qualified windows. The ratio

$$R_k = \frac{\Omega_k^{\mathrm{perturbed}}}{\Omega_k^{\mathrm{reference}}} \tag{26}$$

is a *measured internal-mode ratio*. It is not automatically the physical proper-time factor. The candidate $R_k \stackrel{?}{=} \sqrt{1 - v^2/c^2}$ must be compared only after deriving the dynamics, with full-field Galilean boosts as mandatory null controls.

Euler conserves real-valued circulation set by initial conditions. Integer quantization $\Gamma_n = n\Gamma_0$ requires *additional* microphysics or a phase/order-parameter constraint; it does not follow from Euler circulation conservation alone. Spinors, neutrino mixing, CP/T violation and the Standard Model remain separate unverified closures.

8 Numerical scale check with canonical SST constants

The canonical characteristic circulation and nominal local angular frequencies are

$$\Gamma_c = 2\pi r_c |\mathbf{v}_\circlearrowleft| = 9.68361920349 \times 10^{-9}\ \mathrm{m^2\,s^{-1}}, \tag{27}$$

$$\Omega_c = |\mathbf{v}_\circlearrowleft|/r_c = 7.76344065538 \times 10^{20}\ \mathrm{s^{-1}}, \tag{28}$$

$$|\boldsymbol{\omega}_c| \simeq 2\Omega_c = 1.55268813108 \times 10^{21}\ \mathrm{s^{-1}} \quad \text{only for a locally rigid rotating-core approximation.} \tag{29}$$

With spherical volume $V_c = (4\pi/3)r_c^3 = 1.17163950032 \times 10^{-44}\ \mathrm{m^3}$ and *assuming* homogeneous core density ρ_{core}, one obtains

$$E_{\mathrm{core,\,sphere}} = \frac{1}{2}\rho_{\mathrm{core}}|\mathbf{v}_\circlearrowleft|^2 V_c = 2.72903497979 \times 10^{-14}\ \mathrm{J}, \tag{30}$$

$$\frac{E_{\mathrm{core,\,sphere}}}{c^2} = 3.03646092324 \times 10^{-31}\ \mathrm{kg}. \tag{31}$$

These are diagnostics for a *chosen uniform spherical core*, not dynamically determined energies or measured particle masses. With the background ρ_f, the same hypothetical volume has $E \approx 4.90653 \times 10^{-39}$ J. A density closure is mandatory before switching between these values.

At $\ell = r_c$ and $|\boldsymbol{\omega}_c| = 2|\mathbf{v}_\circlearrowleft|/r_c$, the extra local H^1 energy density $\rho_f \ell^2 |\boldsymbol{\omega}|^2/2$ equals 4 times the ordinary local kinetic energy density. This demonstrates that the regularization is a substantive dynamical model change at core scale, not a negligible correction.

9 Verification contract and decision gates

The accompanying `validate_repair.py` implements **22 analytical/finite-grid checks**: dimensional bases, steady rigid rotation, Euler pressure–Poisson consistency, exact Galilean covariance under uniform full-field boost, H^1 Fourier shear/momentum relation, Euler $\ell \to 0$ limit, a Beltrami ABC helicity check on a periodic grid, and canonical numerical scales. The output is `validation_report.json`. None of these tests substitutes for a time-evolution solver.

Gate	Requirement	Current qualification
G0 Formal algebra	Variational signs, units, dimensions, and limits agree	PASS (analytic checks)
G1 Independent solution	Stationary Euler test + boost null control + pressure Poisson	PASS (symbolic checks)
G2 3D Euler dynamics	3D time integration, $\nabla \cdot \mathbf{v}$, energy/helicity drift, mesh/time convergence	NOT RUN
G3 Source-native knots	PKLSA provenance, knot topology, valid physical cross section, no visual-only relaxation	NOT RUN
G4 Optional H^1 model	Baseline–regularized parity, $\ell \to 0$, convergence and bracket invariants	NOT RUN
G5 Interactions	Independent clock, charge and gravity observables; blind comparisons	UNRESOLVED

Do not promote any matching numerical factor, constructed Coulomb field, local clock ansatz, core pressure gradient, or mass-fitting parameter to a certified physical law at the present stage. This repair does not write to the original paper, Google Drive, the canonical SST library or the frozen falsifier framework.

10 Minimal patch instructions for a future VAM-5 revision

Replace original Section VI eqs. (23)–(26) with Eqs. (7)–(11); original Section VII eqs. (27)–(31) with Euler conservation laws and an explicit "Maxwell/gravitation unresolved" statement; original Section VIII eqs. (32)–(38) with Eq. (17); and original Section IX eqs. (39)–(49) with the distinct vortex-phase-observable protocol above. Move the optional H^1 model to a separate, *model-hypothesis* appendix and eliminate the "quantum, gravitation and electromagnetism derived" language from the title/abstract until independently tested. Keep an unmodified archival version of VAM-5 and issue the revision as a new version with a change log and explicit non-certification status.

Interpretation for a young reader

A toy boat can move with the water or move because the current changes around it. Before calling the boat's motion "gravity" or the turning of a whirlpool "time", we must write one rule that makes the water move correctly and independently measure what the boat and whirlpool actually do.

References

[1] O. Iskandarani (2025), "On a Vortex-Based Lagrangian Unification of Gravity and Electromagnetism: From Knotted Aether Vorticity to Standard Model Reconstruction using Swirlclock Dynamics, Topological Charge, and Emergent Mass," preprint, doi:10.5281/zenodo.15772858.

[2] R. Salmon (1988), "Hamiltonian Fluid Mechanics," *Annual Review of Fluid Mechanics* **20**, 225–256, doi:10.1146/annurev.fl.20.010188.001301.

[3] D. D. Holm, J. E. Marsden, and T. S. Ratiu (1998), "The Euler–Poincaré Equations and Semidirect Products with Applications to Continuum Theories," *Advances in Mathematics* **137**, 1–81, doi:10.1006/aima.1998.1721.

[4] D. D. Holm, J. E. Marsden, and T. S. Ratiu (1998), "Euler–Poincaré Models of Ideal Fluids with Nonlinear Dispersion," *Physical Review Letters* **80**, 4173–4176, doi:10.1103/PhysRevLett.80.4173.

[5] H. K. Moffatt (1969), "The Degree of Knottedness of Tangled Vortex Lines," *Journal of Fluid Mechanics* **35**, 117–129, doi:10.1017/S0022112069000991.